

Class Test

1) a) Data mining:

Data mining is the process of finding useful patterns and knowledge from large amounts of data

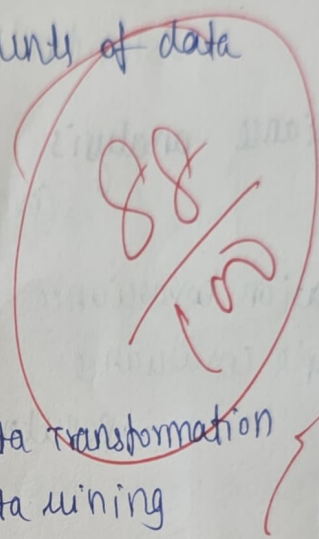
Two Data Mining Tasks:

- Classification
- Clustering

KDD steps:

- Data cleaning
- Data Integration
- Data selection

- Data transformation
- Data mining
- Pattern Evaluation



b) Data cleaning:

Data cleaning is the process of removing errors and improving data quality.

Two Data Quality Issues:

- Missing values
- Noisy data

Handling missing values:

- Replace with mean, median or mode.
- Delete the record if necessary.

Handling Noisy Data:

- Binning
- Smoothing.

2) a) Data integration:  
Data integration combines data from different sources into one dataset

challenges:

- redundancy
- Inconsistency

Handling Them:

- Remove duplicate data.
- standardize data formats.

b) covariance analysis

covariance measures how two variables change together

→ population covariance

→ sample covariance

◦ population covariance

$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$$

◦ sample covariance

$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

Interpretation:

- positive covariance → increase together
- negative covariance → one increases while the other decreases
- zero covariance → independent

3) a)

Given

A

B

2

5

3

8

5

10

4

11

6

14

mean

$$\bar{x} = 4$$

$$\bar{y} = 9.6$$

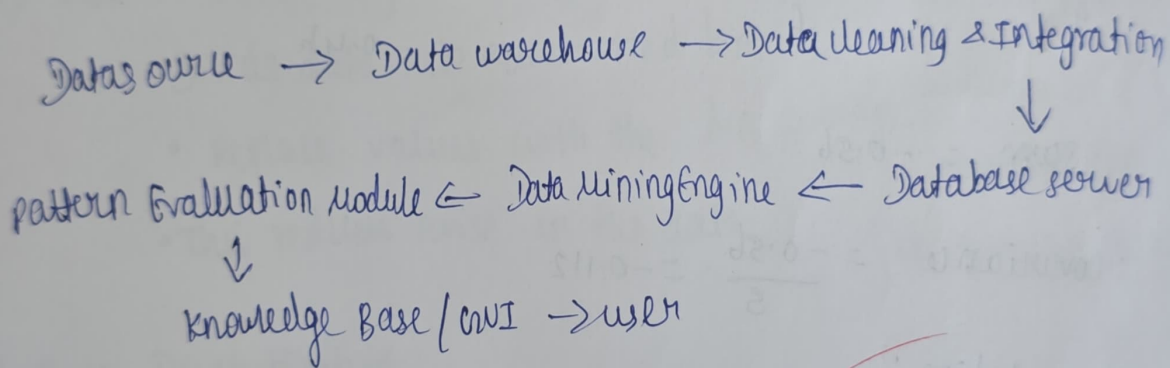
covariance

$$\frac{-4 \cdot 4 + 1 \cdot 6 + 0 + 1 \cdot 4 + 8 \cdot 8}{5} = \frac{16.2}{5} = 3.24$$

covariance = 3.24 (positive)

$\therefore$  stocks A and B rise together  
they are positively related  
they are not independent

b) Architecture of Data mining system



Components

Data sources  $\rightarrow$  Raw data from files and databases

Data warehouse  $\rightarrow$  stores integrated data

Database server  $\rightarrow$  Retrieves required data

Data Mining ~~system~~ engine  $\rightarrow$  performs clustering

Pattern Evaluation  $\rightarrow$  selects interesting patterns

Knowledge Base  $\rightarrow$  Stores domain knowledge

GUI  $\rightarrow$  Displays mining results.



u) a)  $x = (4, 5, 6, 7, 9)$   
 $y = (8, 3, 7, 5, 6)$

Mean

$$\bar{x} = 6.2$$

$$\bar{y} = 5.8$$

| $x_i$ | $y_i$ | $(x_i - \bar{x})(y_i - \bar{y})$ |
|-------|-------|----------------------------------|
| 4     | 8     | -4.84                            |
| 5     | 3     | -5.04                            |
| 6     | 7     | -0.24                            |
| 7     | 5     | -0.46                            |
| 9     | 6     | 0.44                             |

Sum = -0.56

$$\text{Covariance} = \frac{-0.56}{5} = -0.112$$

Interpretation:

\* there is a weak negative relationship

b) Major Data processing steps

- Data cleaning
- Data integration
- Data transformation
- Data reduction
- Data discretization

Proposed pipeline

1 → Remove missing values



Remove outliers

Standardize formats



normalize data



Reduce data



store clean dataset

Justification

- cleaning first removes errors
- normalization improves mining
- reduction decreases storage and computation

5) a) Equal-Frequency Binning:

- Divide data into equal-sized bins
- Replace values with the bin average
- This reduces noise in the data

Bin: 1

13, 15, 16, 16, 19, 20, 20,  
21, 22

mean = 18

smoothed Bin

18, 18, 18, 18, 18, 18, 18, 18, 18

b) Data Discretization:

It converts continuous data into intervals.

Importance

- Reduce data size
- Remove noise
- Improve classification accuracy

Equal-width Binning

range divided into equal widths.

Impact on Data Analysis

- Equal-width is easy but may create unbalanced bins.